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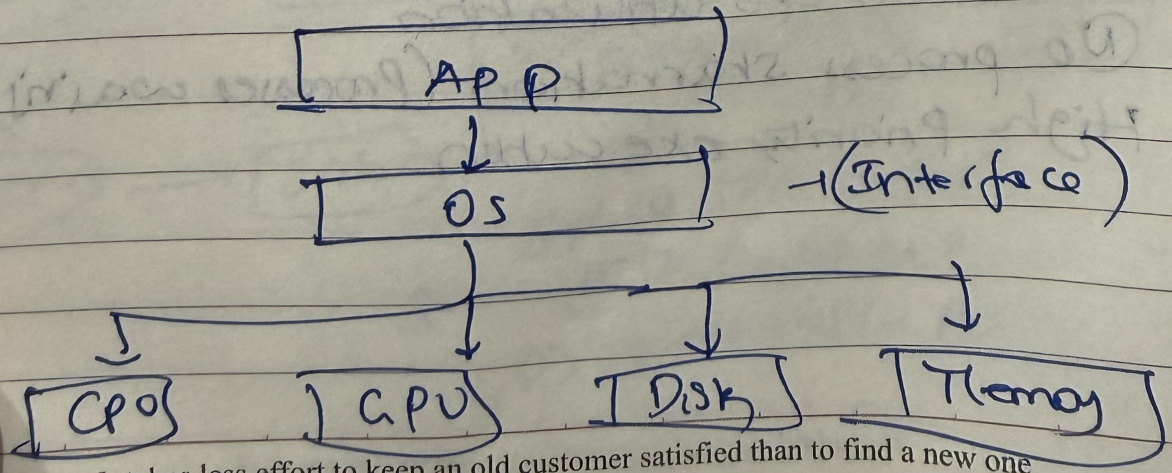
# Operating System Fundamentals

If We are using an application, e.g) Tiktok, it will take up all the Resources:

* CPU * Memory
* GPU * Disk

(Now, no other app will be able to use it, this is called Resource Hacking.

But, what if there's a layer in b/w the Application & these Resources? It can manage all the Resources properly & allocate it as per the app's need. That layer is OS.



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→ OS completely takes care of the Resource Management & Memory Management, so that, the App dev can only focus on writing the Apps rather than thinking about How to manage Resources & Memory.

→ Isolation & Security:

OS provides isolation & security to different application by splitting them up properly & not allowing one app to interfere with the Resource/Memory Management of another App.

Main Goals of OS:

→ Maximum CPU utilization
→ No process starvation (Processes waiting)
→ High Priority execution

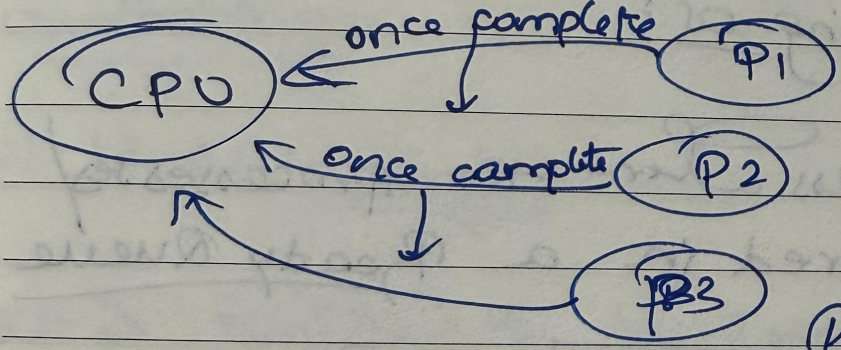
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# Types of OS

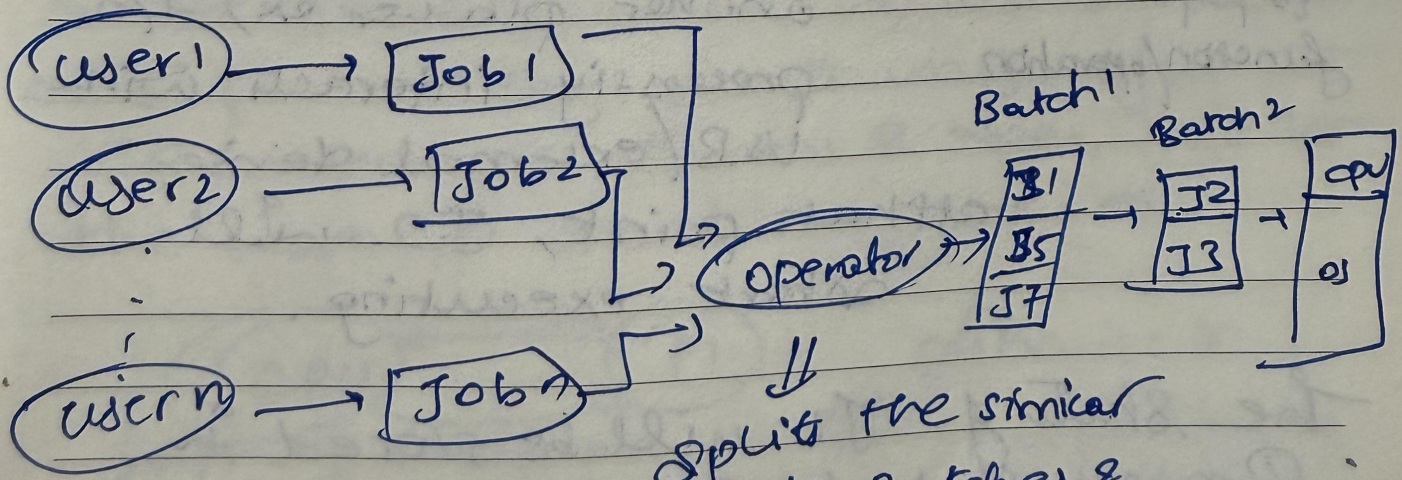
## (i) Single Process OS:

Sequential processing, one By one.




Disadvantage if one Sequential process and other Processes have to wait until the whole process complete

## (ii) Batch Processing



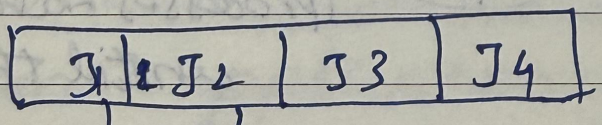

Split the similar jobs into Batches & sequentially process

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But here also, when J1 & J4 go together as batches, while J1 executing, J4 has to wait. So, ① No maximum cpu utilization ② No proper Resource Alloc

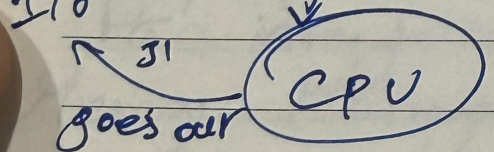
(iii) Multiprogramming os:

It is the process where the processes/jobs are stored in a Ready Queue




⇒ Ready queue

J2 start exec




goes out to perform function/operation

When J1 goes out to another place, ex) I/O processing interact with USB/external device, at that point, J2 will start executing.

The state of J1 will be stored in Process Control Block (PCB)

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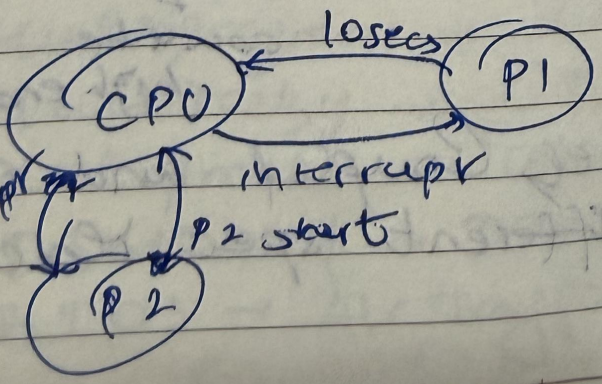
when again J1 needs to be executed, ie, once it reaches back from I/O to CPU, CPU fetches its state from PCB.

This whole process is called Context Switching.

(iv) Multitasking OS:

It is similar to Multiprogramming OS, with one advantage, which is Time sharing.

Say, we have 3 processes, they have to be executed at same time, so, the CPU has a set Time Quantum, ex) 10secs. For 10secs one process can do its work, after that CPU interrupts, Process B does, like not the time is shared among the OS.




⇒ Likewise it keeps sharing the time allocation.

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(v) Multiprocessing:

→ Has context Switching
→ Has Time Sharing
→ Add on Advantage ⇒ More than 1 CPU

⇒ processes will be properly scheduled b/w CPUs.

(vi) Distributed OS:

→ It can handle multiple CPUs & Memory at the same time, even if they are of different Machines.

Ex) CPU Mem → could be low end → OS → CPU Mem
CPU Mem
CPU Mem → could be high end

Everything is loosely Coupled and OS can handle different spec Machines at same time.

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## (vii) RTOS (Real Time OS)

→ It is a working Real time OS, which will have near zero error percentage.

⇓

No Errors must take place & the computation must be highly efficient. (ex) very low latency.

Ex) In Flight systems / Nuclear systems / Industrial systems

⇓

No chance of Errors.

**Examples:**

(i) Single process → MS DOS
(ii) Batch process → ATLAS
(iii) Multiprogram → THE
(iv) Multitasking → CTSS
(v) Multiprocess → Windows.
(vi) Distributed → Interconnected Computers.
(vii) RTOS → Air Traffic control system.

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# Multi Threading:

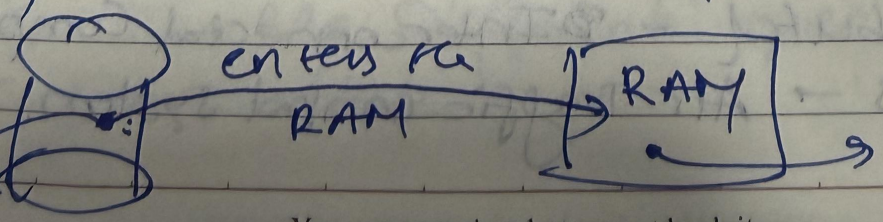
## What's a Process?

So, initially, an Application would be in code format, stored in a file format like .cpp (for c# prog).

Now, in order to make this executable, it has to be compiled. So, the code file converts to executable file.

.cpp → .exe

Now, this .exe file stored inside a Disk in our computer. Now, to run this, it has to be taken out from Disk, and executed in Ram. Once a ~~process~~ ~~enters~~ program enters the RAM from Disk, it becomes a process



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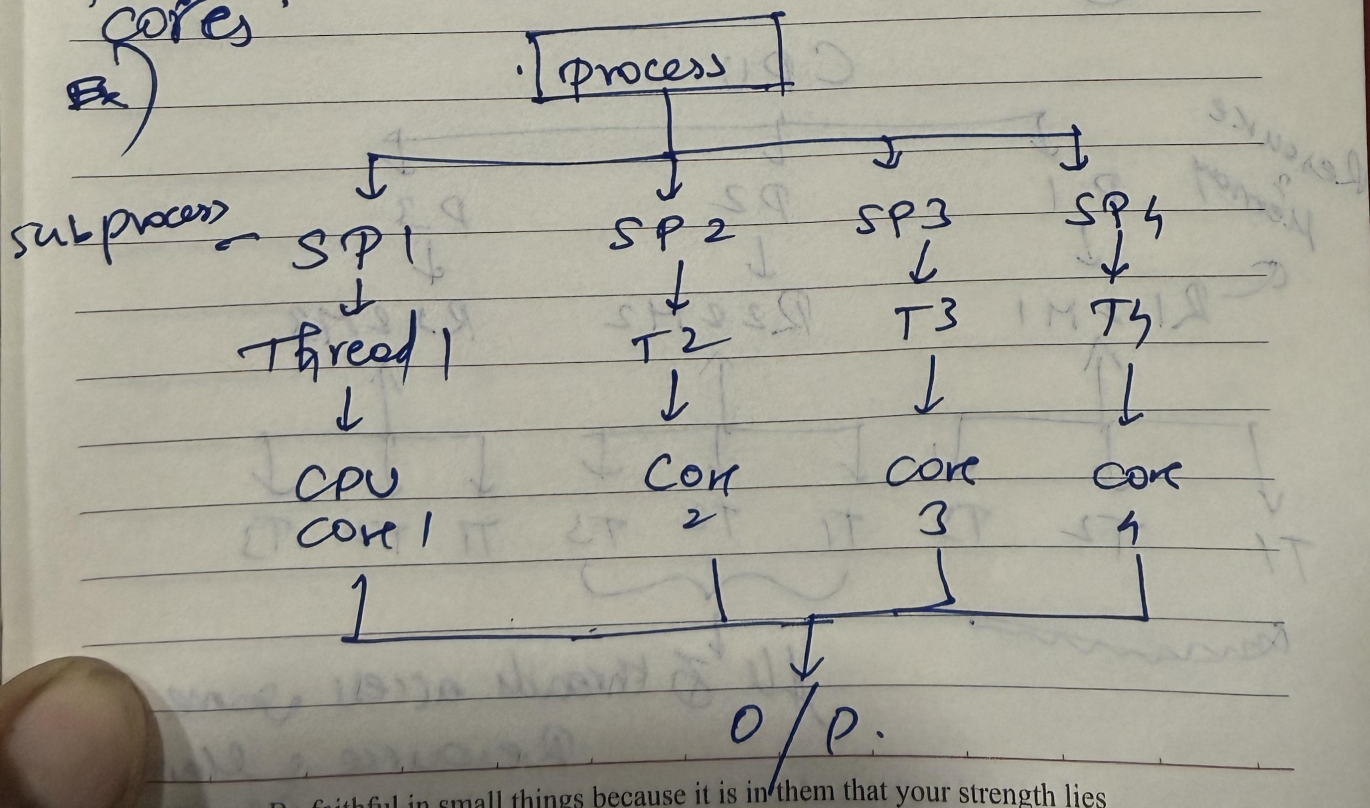
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# what is a Thread?

Thread is an executable unit of a Process, ie, A large process could be split up & given to respective threads which can execute it parallely.

In a CPU, this Data parallelism happens via different cores. As Multiprocessig OS contains more than one CPU cores, each thread could be processed parallely across these cores.

Ex)



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# Problem in Multithreading:

It has no concept of Isolation & Mem. protection.

It is because, All threads belong to a single process, and, CPU would have Allocated certain amt of Memory & Resources for that process. Hence all threads need to access the same Resources & Memory. Tht could effect the isolation & Memory Management.

